MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5560

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, mercury

See figure 1

Mounting position: Vertical base down

Weight: 6.5 ounces (184.3 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	ib	Ib	igl	ig2	THg	Ec2	tk
Unit:	Vac	v	v	a	Adc	a	a	°C	Vdc	sec
Maximum:	5.25	1,000	1,000	15.0	2.5	1.0	1.0	75	---	---
Minimum:	4.75	---	---	---	---	---	---	40	---	300
TEST CONDITIONS:	5.0	---	---	---	---	---	---	38 to 42	0	300

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	4.3	4.9	Aac
3201	Critical grid voltage for conduction (1)	-	Ebb = 100 Vdc; Rp = 40 to 5,000 ohms; Rgl = 0	0.65	II	Ecl	-4.0	+6.0	Vdc
3201	Critical grid voltage for conduction (2)	-	Ebb = 1,000 Vdc; Rp = 400 to 5,000 ohms; Rgl = 0	0.65	II	Ecl	-5.0	-13.0	Vdc
1231	Peak emission by voltage drop	3,6	ib = 15a	0.65	II	etd	---	20	v
	Quality conformance inspection, part 2								
3216	Grid current	5	Epp = 220 Vac; Ib = 2.5 Adc; Ef = 5.5 Vac; Rgl = 100 Meg	---	---	Ic	---	0.5	μAdc
(B) ---	Operation (1)	1,4	epx = epy = 1,100 v (min); Rgl = 10 Meg; Egl = 120 Vac (max); Ib = 2.5 Adc	---	---	---	---	---	---
1031	Vibration	-	No voltages applied	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group D; Epp = 110 Vac; Ib = 2.5 Adc; (min); ib = 15a; Rgl = 1 Meg; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	24	v
3201	Critical grid voltage for conduction (2)	-		---	---	Ecl	-4.0	14.0	Vdc
(B) ---	Operation (2)	1,2,7	Operation (1)	---	---	---	---	---	---

NOTES:

1. With rated heater voltage, the tube shall be operated in a rectifier circuit with a resistance load and without a filter in the load circuit. The overall regulation of the anode supply voltage shall be less than 10 percent. A grid-to-cathode capacitor may be used to reduce voltage surges on the grid. After the mercury has been properly distributed, the tube shall be operated for 5 minutes at specified conditions without injury or sign of arc-backs or sputter, and at the end of the period shall show uniform control characteristics when the phase of the grid voltage is varied in relation to the anode voltage.
2. Two tests shall be made, one with condensed-mercury temperature not greater than 40°C and the other with condensed-mercury temperature not less than 75°C.
3. The grid shall be connected to the anode through a 10,000-ohm resistance.
4. The ambient temperature shall be between 20° and 40°C and artificial cooling shall not be used.
5. The tube shall be operated for 5 minutes immediately prior to test. Ambient temperature shall be 20° to 40°C, with no control of mercury temperature.
6. This test is to be the first test performed at the conclusion of the holding period.

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NOTES: -Continued

Ⓑ 7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection (QCI), part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

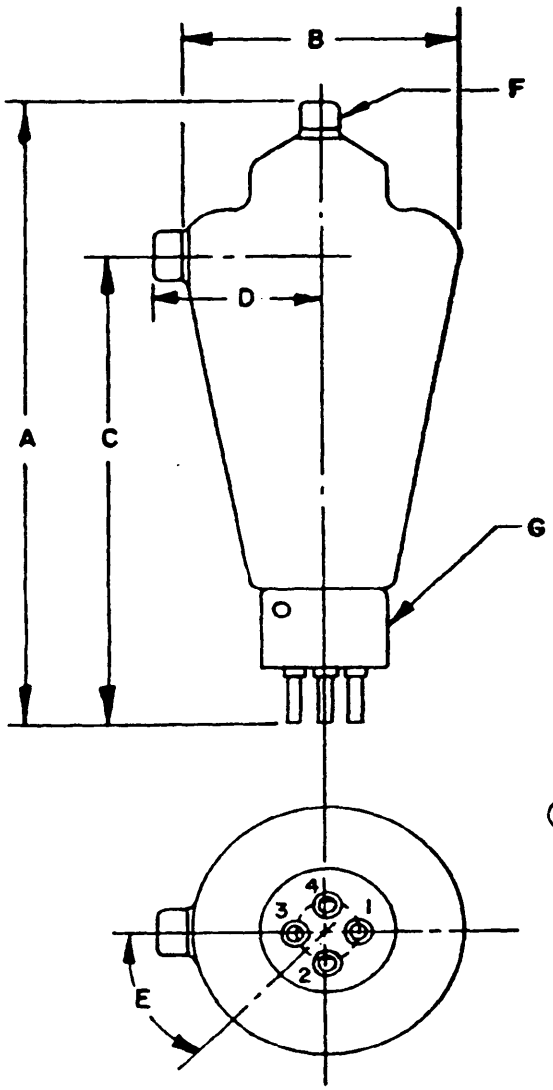
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2919)

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PIN CONNECTIONS	
Pin No.	Element
1	h
2	k
3	g2
4	h & k
Top cap	a
Side cap	g1

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	7.438 (188.93)	7.938 (211.63)
B	---	3.000 (76.20)
C	5.625 (142.88)	5.875 (149.23)
D	---	2.250 (57.15)
(B) Quality conformance inspection, part 3 (see note)		
E	40°	50°
F	Cap: C1-5 (EIA)	
G	Base: A4-10 (EIA)	

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 5560.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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